

```
align(center + horizon)[ block( width: 15cm, inset: (y: 1.1cm), stroke: (top: 2.5pt + rgb(26, 54, 93),
bottom: 0.75pt + rgb(203, 213, 225)), ) [ align(center) [ text(size: 10.5pt, tracking: 0.35em, fill:
rgb(71, 85, 105)) [大 作 业] v(0.55cm) text(size: 26pt, weight: "bold", fill: rgb(15, 23, 42)) [揭棋对弈
程序设计] v(0.65cm) text(size: 19pt, weight: "medium", fill: rgb(30, 64, 175)) [领域模型] v(0.35cm)
if subtitle != "" [ text(size: 13pt, fill: rgb(100, 116, 139)) [Domain Model — 核心类设计、状态机、
不变式] v(0.35cm) ] ] ]
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v(1.5cm)
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box( width: 13cm, inset: (x: 1.2cm, y: 0.95cm), fill: rgb(248, 250, 252), radius: 6pt, stroke: 0.75pt +
rgb(226, 232, 240), ) [ align(center) [ text(size: 11pt, weight: "bold", fill: rgb(51, 65, 85)) [小组成员]
v(0.55cm) grid( columns: (1fr, 1fr), column-gutter: 1.6cm, row-gutter: 0.65cm, align: center +
horizon, [text(size: 13pt, weight: "bold") [张恒基 (组长) ]
text(size: 12pt, fill: rgb(100, 116, 139)) [2024211301 / 2024210926]], [text(size: 13pt, weight: "bold")
[秦博宇]
text(size: 12pt, fill: rgb(100, 116, 139)) [2024211302 / 2024210940]], [text(size: 13pt, weight: "bold")
[陈艺博]
text(size: 12pt, fill: rgb(100, 116, 139)) [2024211302 / 2024210931]], [text(size: 13pt, weight: "bold")
[陈雨飞]
text(size: 12pt, fill: rgb(100, 116, 139)) [2024211301 / 2024210918]], ) ] ]
```

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v(1cm) text(size: 12pt, fill: rgb(100, 116, 139)) [技术设计 · 2026-06-18] v(0.5cm) text(size: 10.5pt,
fill: rgb(148, 163, 184)) [北京邮电大学 · 计算机学院] ]
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核心领域类

类	包	职责
Board	core	10×9 棋盘，棋子查询，move/undo
ChessPiece	core	棋子状态：color, type, virtualType, revealed
Move	core	走法数据：source, destination
Game	core	对局状态机：board, currentTurn, status
GameRecord	record	文字棋谱：lines: List<String>
GameRecordStore	server	棋谱落盘：records/*.jieqi
ReplayTimeline	record	复盘时间线：frames: List<ReplayFrame>
ReplayFrame	record	复盘单帧：stepIndex, move, boardSnapshot
EndgameJudge	core	终局判定：checkAfterMove
RuleValidator	core	走法生成与合法性校验

类关系

Game 1 → 1 Board
Game 1 → 1 GameRecord
Game 1 → 1 ReplayTimeline
Board 1 → * ChessPiece
Game → EndgameJudge
Game → RuleValidator

状态机

状态转换	触发条件
WAITING → PLAYING	双方 ready + 协商先手
PLAYING → RED_WIN	黑将被吃 / 黑被将死 / 黑困毙 / 黑长将/长捉 6 次 / 黑超时
PLAYING → BLACK_WIN	红帅被吃 / 红被将死 / 红困毙 / 红长将/长捉 6 次 / 红超时
PLAYING → DRAW	40 步无吃子 / 兵卒长捉和 / 双方同意和棋
任意 → TIMEOUT	当前回合超时 65s

不变式约束

约束	说明
棋盘独占	红黑双方永远不共享同一个格子
回合唯一	currentTurn 只能是 RED 或 BLACK
暗子约束	revealed=false 时 type 可被翻子改动，但 virtualType 永不变
步数上限	noCaptureCount ≤ 80
坐标范围	row 0-9, col 0-8, 九宫为 row 7-9/0-2, col 3-5